Capstone Project Case Study – Book-My-Show DevOps Lifecycle

This capstone project is designed to simulate a real-world DevOps workflow for the Book-My-Show (BMS) application. Candidates are required to follow the complete CI/CD lifecycle – from code collaboration to deployment and monitoring. Optional integrations like Trivy and OWASP Dependency Check can be added for advanced security scanning.

Step 1: Jira Workflow

• Each candidate must assign themselves tasks in Jira.
• Track task progress using Jira board (To Do → In Progress → Done).
• Export the board as part of the final submission.

Step 2: GitHub Workflow

• Clone the GitHub repository: https://github.com/akshu20791/Book-My-Show/
• Create a feature branch and make changes.
• Push changes and raise a Pull Request (PR).
• Review peers' PRs and approve before merging into the main branch.
• Submit the final PR link as deliverable.

Step 3: Jenkins CI/CD Pipeline

Pipeline stages to implement:
1. Clean Workspace
2. Checkout Code from GitHub
3. SonarQube Analysis (Quality Gate)
4. Install Dependencies (NPM)
5. Trivy FS Scan (Optional)
6. OWASP Dependency Check (Optional)
7. Docker Build & Push to DockerHub (via Jenkins)
8. Deploy to Docker Container
9. (Optional) Deploy to Kubernetes (EKS)
10. Email Notification on build result

⚠ Note: Candidates must implement this pipeline using a Jenkinsfile and submit the Jenkinsfile as a deliverable.

Step 4: Docker Deployment

• Candidates must write their own Dockerfile to build the BMS app image.
• Build Docker image and push to DockerHub via Jenkins.
• Run container locally and validate accessibility on port 3000.

Step 5: Kubernetes Deployment (EKS)

• Candidates must write their own Kubernetes manifests (`deployment.yaml`, `service.yaml`).
• Deploy the application on EKS cluster.

• Expose service using NodePort or LoadBalancer.
• Validate deployment using 'kubectl get pods' and 'kubectl get svc'.

Step 6: Monitoring & Observability

• Install Prometheus and Node Exporter to collect metrics.
• Integrate Jenkins metrics into Prometheus.
• Install Grafana, configure Prometheus as a data source.
• Add dashboards for Node health and Jenkins performance.
• Submit Grafana screenshots in deliverables.

Final Deliverables

1. Jira Board Export (task progress).
2. GitHub PR link.
3. Jenkins pipeline execution logs.
4. Jenkinsfile (mandatory).
5. SonarQube Quality Gate report.
6. (Optional) Trivy/OWASP reports.
7. DockerHub repository link.
8. Dockerfile (mandatory).
9. EKS deployment proof (kubectl outputs).
10. Kubernetes manifest files – deployment.yaml and service.yaml (mandatory).
11. Prometheus & Grafana screenshots.
12. Email notification screenshot.
13. Final summary report (Word/PPT).

Flowcard to Follow

Jira → GitHub Branching & PR → Jenkins (Jenkinsfile) → Code Analysis (SonarQube, Trivy*, OWASP*) → Docker Build & Push (Jenkinsfile) → Deploy (Docker / Kubernetes with manifests) → Email Notification → Monitoring (Prometheus & Grafana)
*Optional steps for security scanning

Additional Mandatory Deliverables (Reviewer Feedback)

Based on SME review, the following items are mandatory deliverables for candidates:

1. Jenkinsfile – Candidates must implement the CI/CD pipeline using a Jenkinsfile. The Jenkinsfile itself must be submitted as part of deliverables.

2. Docker Push via Jenkins – Docker image push to DockerHub must be done through the Jenkins pipeline (not manually).

3. Dockerfile – Candidates must write their own Dockerfile to build the application image. The Dockerfile should be included in deliverables.

4. Kubernetes Manifests – Candidates must write their own Kubernetes YAML files (`deployment.yaml`, `service.yaml`). These manifest files should be submitted as deliverables.